

Knowledge Graphs and Ontologies: Beyond the Dictionary Fallacy

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Dr Nicolas Figay

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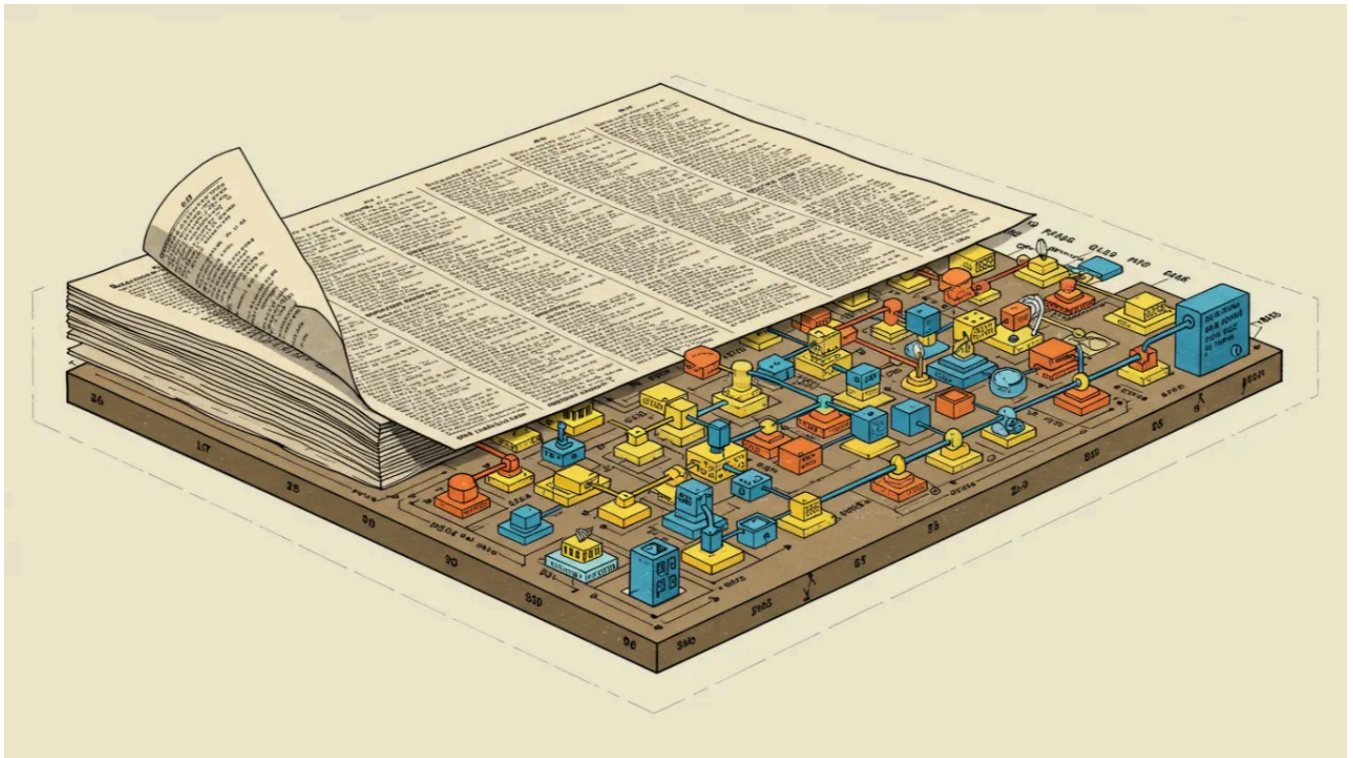
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Most knowledge graph practitioners treat ontologies as sophisticated dictionaries — structured vocabularies and entity hierarchies optimized for data retrieval and computational efficiency. This pragmatic approach, while useful for engineering data systems, misses something essential about what ontologies truly are. Crucially, it prevents us from leveraging their full power as instruments of collective understanding and coordinated action.

The Confusion of Structure and Being

The reduction of ontology to schema stems from a fundamental confusion. An ontology, in its philosophical and formal sense, is not merely an inventory of what exists; it is the science of being itself — a rigorous formalization of the fundamental categories and relationships through which a domain makes sense. It seeks to answer: *What is the essential, coherent structure of this domain?*

Yet, in the context of data architecture, knowledge graphs reduce ontologies to simple schema layers: entity types, properties, hierarchies, and lightweight axioms that primarily serve machine learning and query resolution. This is like confusing a grammar rulebook with literature itself; one enables construction, but the other formalizes the deep, meaningful structure that drives interpretation and communication.

The Missing Piece: Community Epistemology

The critical insight that moves us beyond this fallacy is that ontologies are not discovered — they are constructed by communities.

Knowledge is never solely the result of isolated perception. It emerges through a dynamic movement from the individual (tacit knowledge, context-bound intuition) to the collective (shared formalization, domain expertise, collective validation). An ontology is the formal expression of this shared understanding — stabilized, teachable, independent of any single practitioner, yet always situated within a specific domain.

This collective negotiation establishes an epistemic community's core commitments. It recognizes that knowledge is:

- Situated within a domain and its community of practice.
- Constructed through collective negotiation and validation.
- Formalized in ways that enable coherent communication and computation.
- Durable because it transcends individual perspective.

The Architectural Challenge: Pluralism and Upper Ontologies

The ISO 42010 standard for architecture description teaches a lesson the knowledge graph community must heed: a single, flattened schema is fundamentally insufficient for complexity. Architectural descriptions require multiple viewpoints — each valid, necessary, yet each representing a distinct concern (e.g., security,

performance, cost). This necessary pluralism cannot be collapsed into a single graph.

This necessity for managing pluralism is precisely where Enterprise Architecture Modeling (EAM) and Architectural Description Languages (ADLs) such as ArchiMate become crucial. EAM shifts the focus from purely technical data structure to the holistic organizational context. It provides the framework for describing the working enterprise — its business capabilities, applications, processes, and technology infrastructure — thus ensuring that the conceptual commitments of the upper ontology are contextualized within the operational reality where work and action occur with the necessary directions, guidelines, and supports.

This is where the Upper Ontology becomes critical. It is not something to simply reuse wholesale, but rather the essential top conceptual layer that enables the harmonization and alignment of diverse, domain-specific ontologies.

An upper ontology answers foundational questions: *How do we distinguish entities from events? Objects from processes? Roles from agents?* Crucially, it makes explicit the philosophical commitments about what kinds of things can fundamentally exist and how they relate. It is the conceptual infrastructure that allows different domain perspectives (viewpoints) to be mapped, translated, and integrated coherently.

This work is design, not passive reuse. It requires making intentional choices about the fundamental structure of the domain, grounded in how that domain *actually functions*. Without this rigorous upper layer, different viewpoints collapse into incompatible silos; with it, they become coherent perspectives on a unified reality that supports both local autonomy and global integration.

Toward a Philosophy for Action

The ultimate failure of the “dictionary fallacy” is that it forces us to formalize the world only in terms of static description. To be truly adequate, ontologies must be grounded in a philosophy of action.

An adequate ontology must formalize not merely what exists, but how a domain functions — what agents can do, what constraints govern possibility, what purposes drive choices, and how knowledge enables or limits what is possible. This is not an extra layer; it is the substance of the ontology itself.

This demands formalizing key non-structural dimensions:

1. **Intentionality:** What purposes does this knowledge serve? For whom is this formalization constructed?
2. **Normativity:** Capturing not just what *is*, but what *should be* — what is valid, prohibited, or required (rules, policies, ethics).
3. **Agency:** Explicitly defining who acts (agents), their capacities, responsibilities, and constraints.
4. **Temporal and Process Dynamics:** Formalizing how states change, what processes unfold, and the sequences of events that matter.
5. **Justification Structures:** The reasoning, evidence, and criteria that guide justified decision-making and action within the domain.

These intentional, normative, and dynamic aspects cannot be adequately reduced to graph structure and logical predicates without significant loss of context and meaning.

Implications for Knowledge Graph Design

If we accept this richer, action-oriented view, the mandate for knowledge graph design shifts:

1. **Design from Bedrock:** Ontologies must be grounded in rigorous upper ontologies that provide the foundational conceptual architecture for the domain — not as abstract formalism, but as a reflection of how action and change truly work within it.
2. **Embrace Pluralism:** The upper ontology must enable and manage multiple, legitimate viewpoints (domain schemas, different formalizations) to coexist coherently, validating the ISO 42010 insight that architectural pluralism is a necessity, not a weakness.
3. **Epistemology is the Work:** The real effort in ontology design is not engineering a taxonomy, but engaging deeply with communities of practice to formalize not just their vocabulary, but their logic of justified decision-making and collective change.

This fundamental reorientation — from treating ontologies as mere data infrastructure to seeing them as philosophical frameworks for collective

understanding and coordinated action — is what ultimately transforms knowledge graphs into genuine instruments of complex organizational function.

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Biases And Fallacies



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What are your thoughts?



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Oct 21, 2025



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Semantichasm

Oct 24, 2025



Absolutely right!

That way of thinking is outdated and was mainly driven by the development of the internet - as a result now most graph DBs are half baked ontologies supporting reasoning systems with data and no consideration for coherence in... [more](#)



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Alex Kurasoff

Nov 6, 2025



Excellent article! And, beware of intentionally "overfitting".



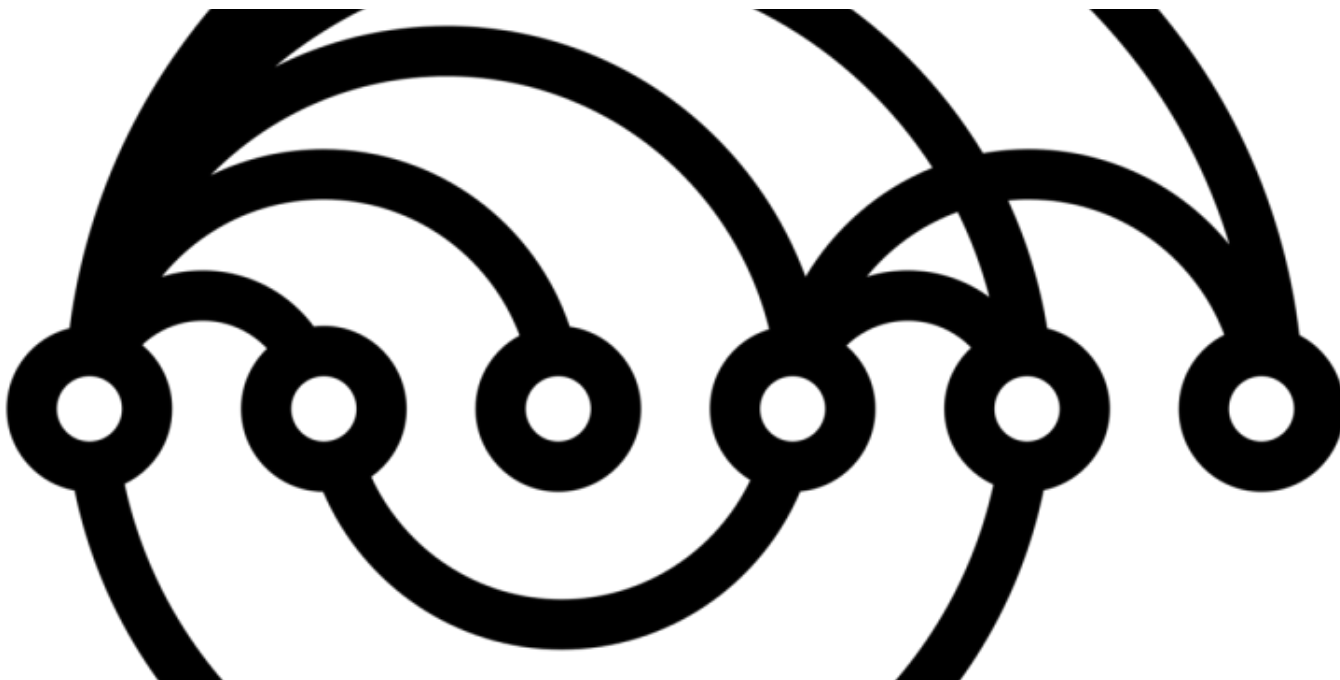
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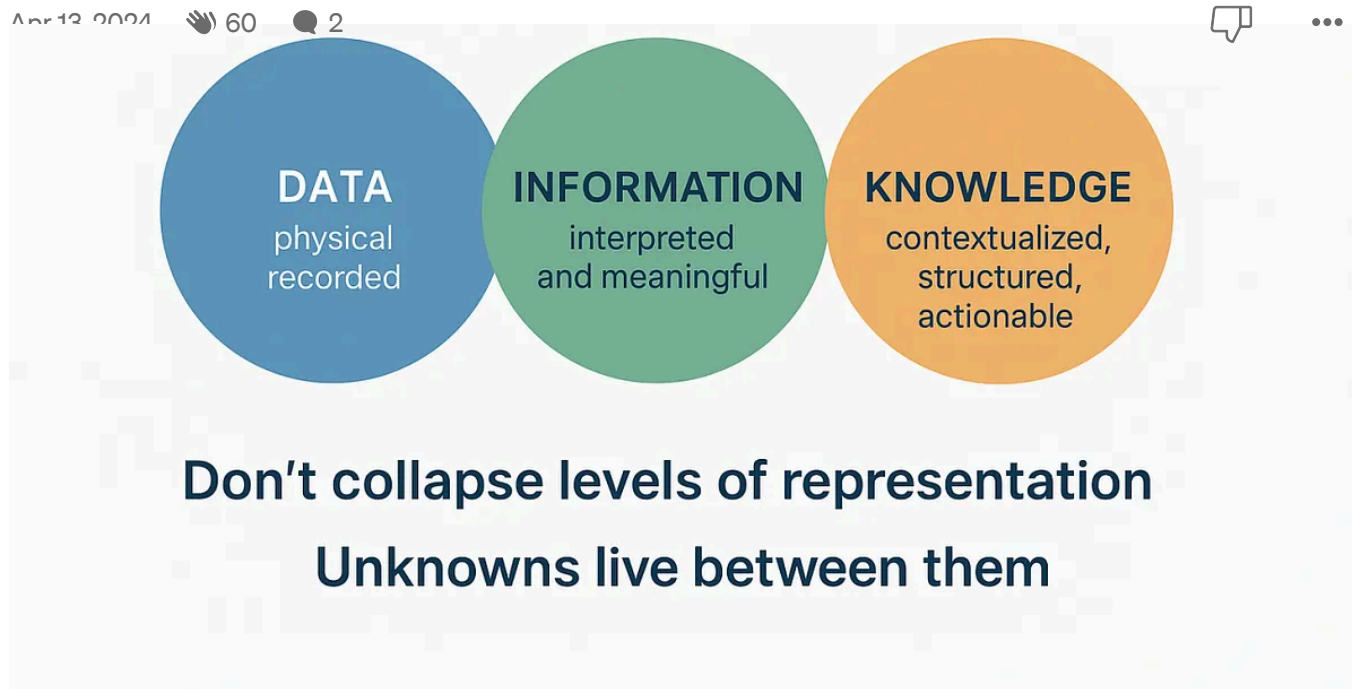
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What is an ontology in the Artificial Intelligence context

Definition

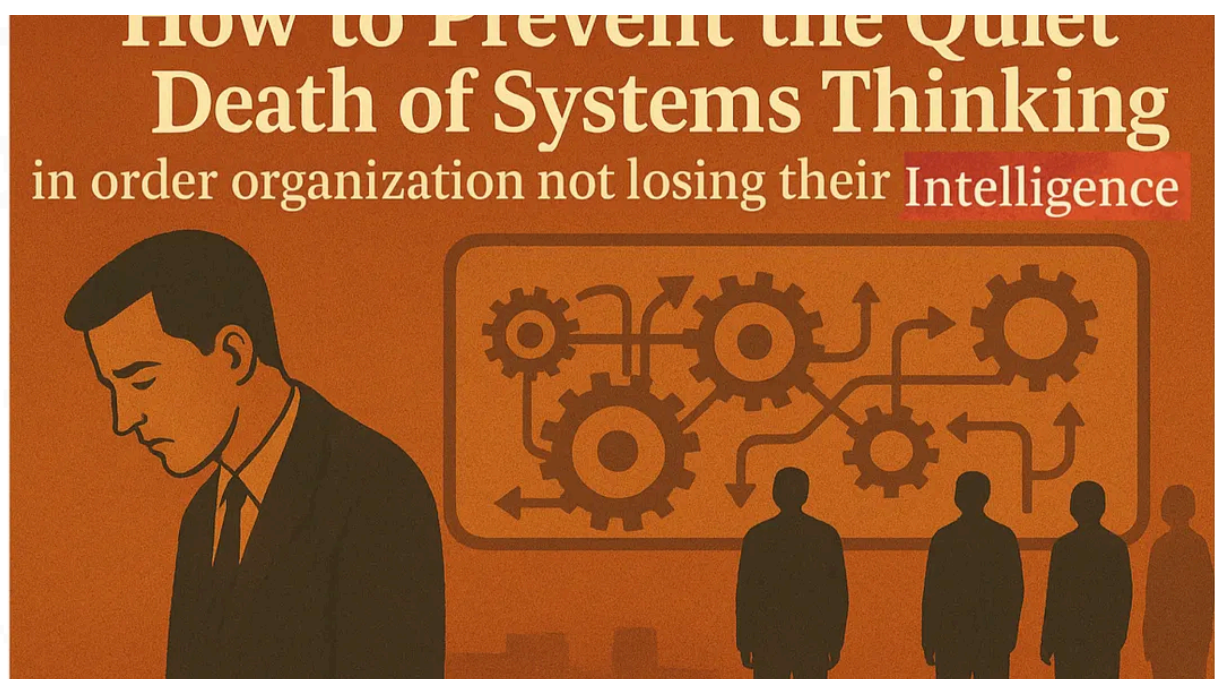


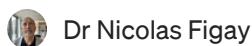
 Dr Nicolas Figay

When Knowledge Graphs Fail, It's Not the Ontology — It's the Epistemology*

Let's have a look to the glossary at the end of the article which explains the concepts and technologies used in this article.

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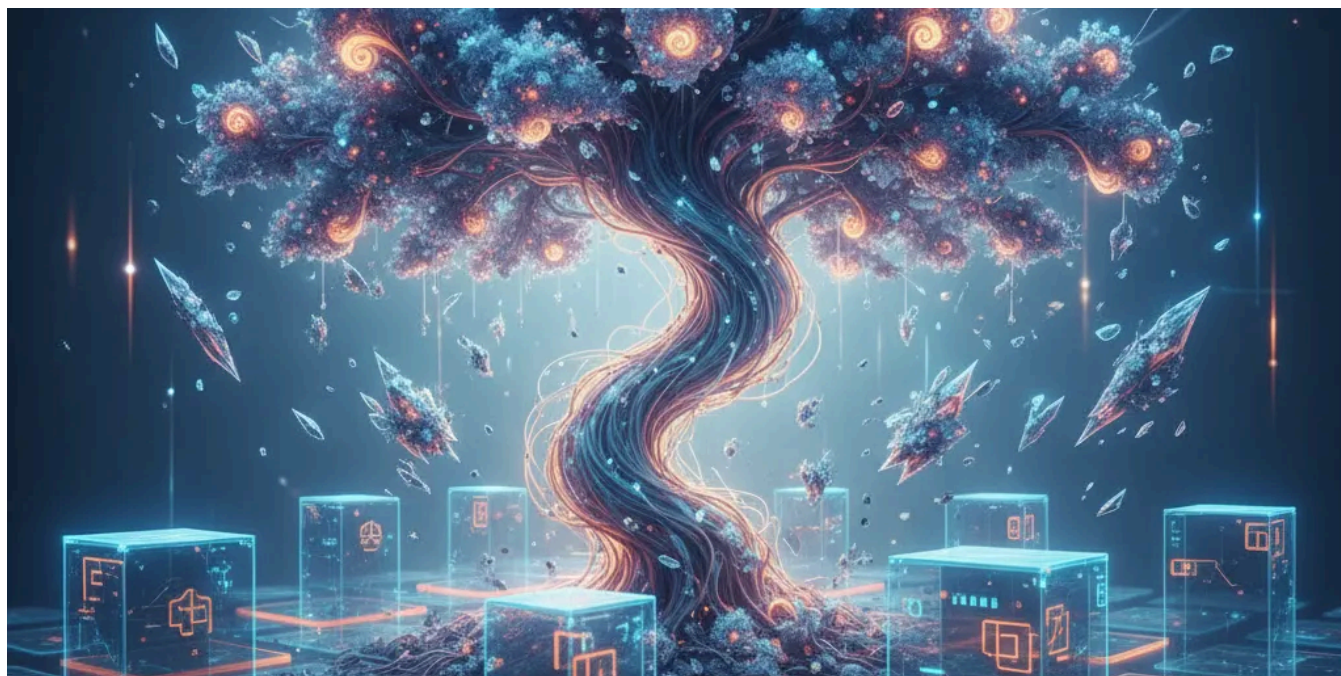




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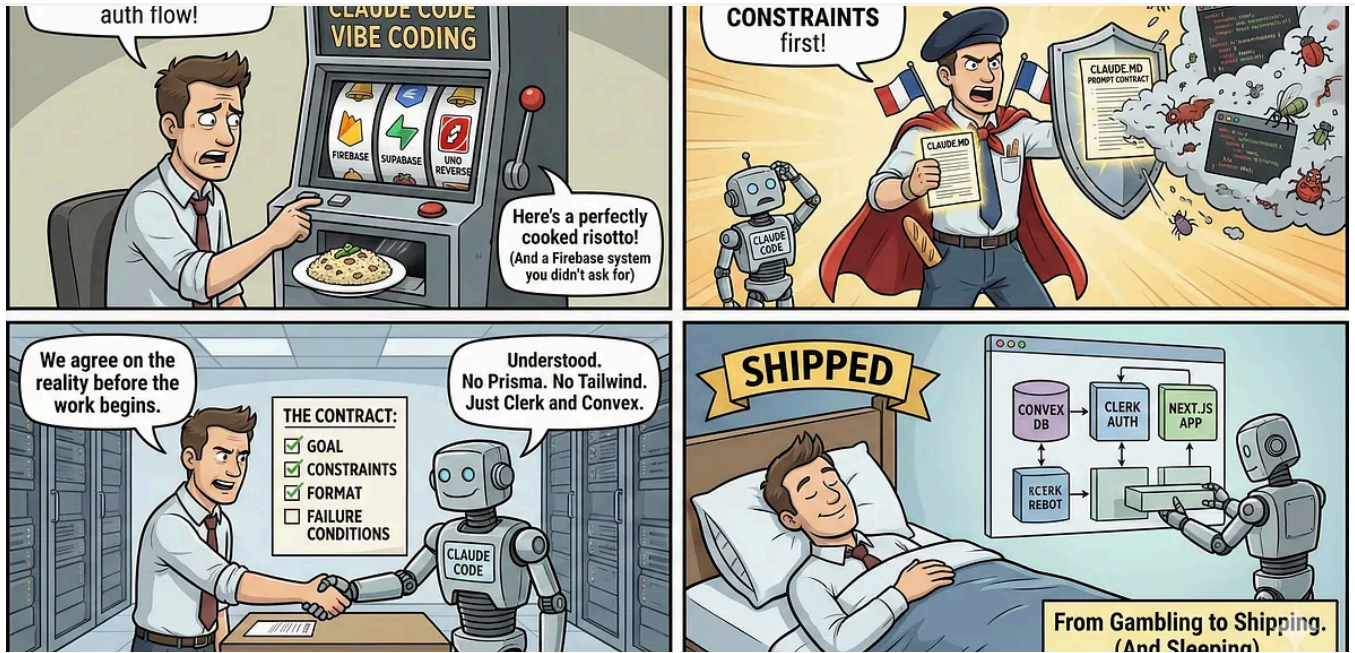


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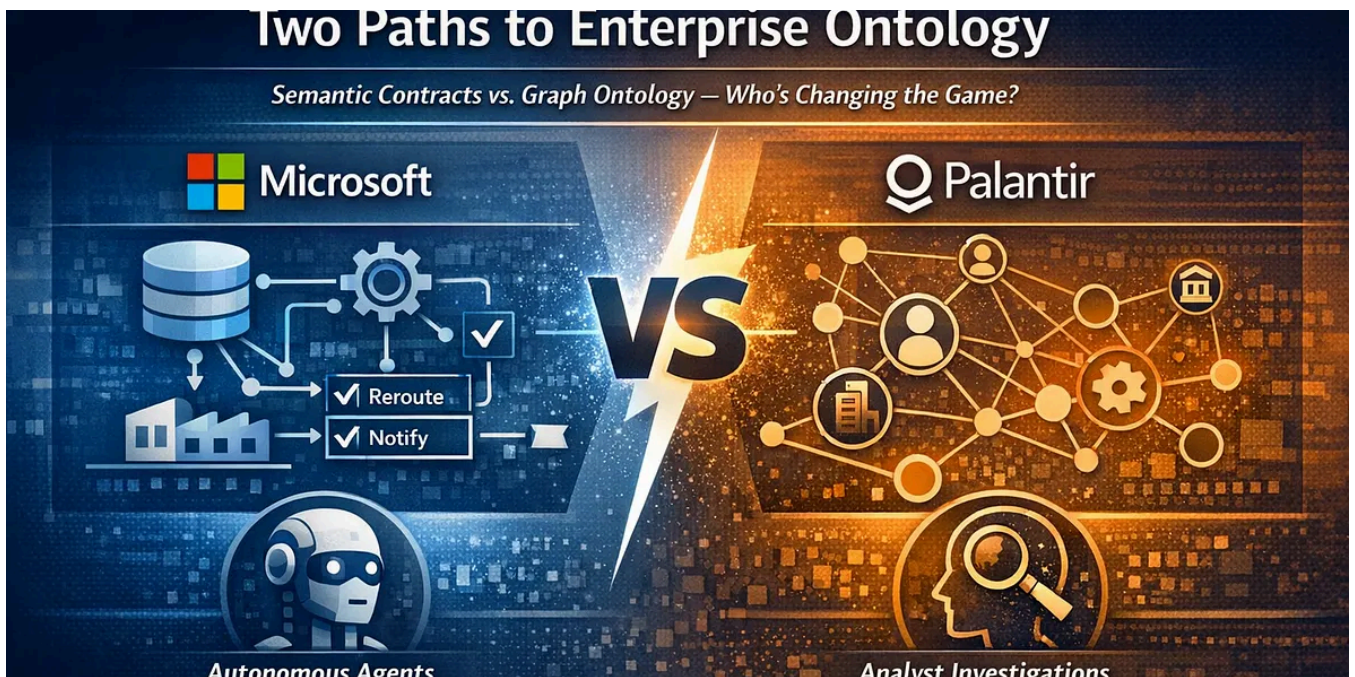


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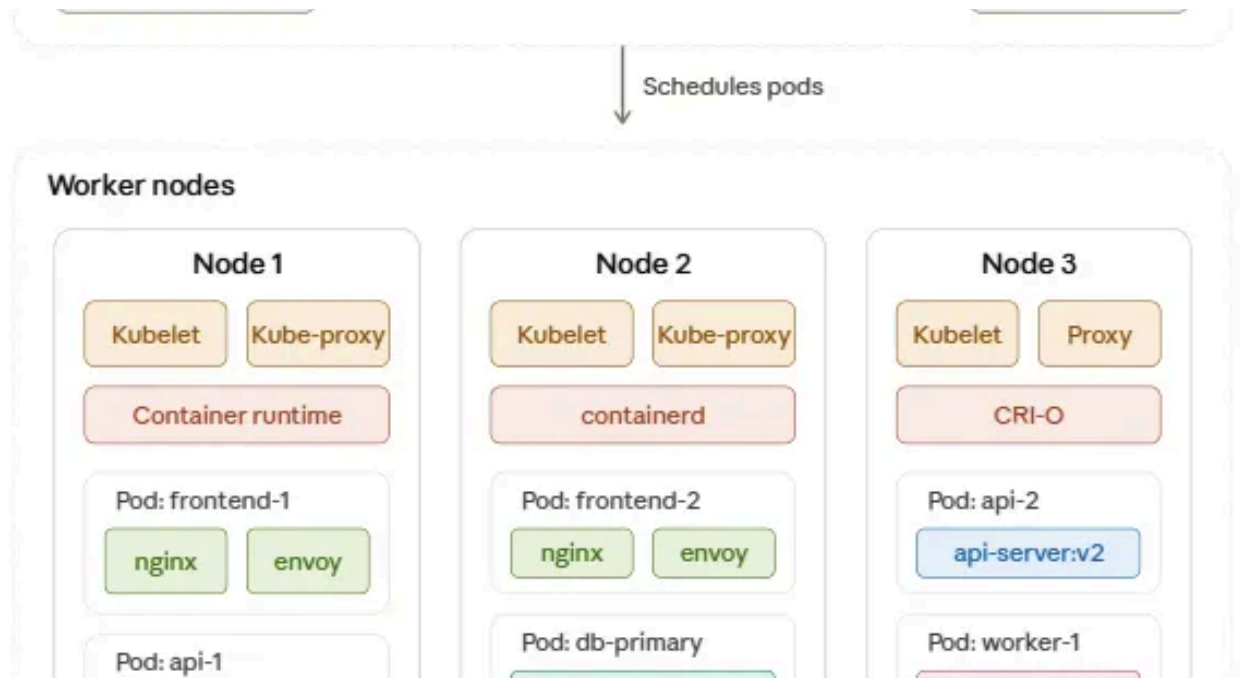


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